

Dish Promotion Score

A formula that scores each dish so the AI promotes the most profitable one

1. What it does

Every field is in a different unit (rupees, pieces, minutes, flags), so they can't simply be added. Two steps fix that: turn the fields into one comparable scale, then combine them with weights into a single **score centred at zero** — a positive score means *promote this dish*, a negative score means *don't*. The dish with the highest score is the best one to push right now.

2. Put every field on the same scale

Profit proxy — value density. Until cost is added, the best stand-in for profit from price and quantity is how much money a dish earns per unit of food served:

$$V_i = \text{Price}_i \div \text{Quantity}_i$$

more rupees for a smaller portion = higher margin (e.g. Dosa ■95÷1 beats Idly ■50÷2)

Standardise the value density (V) and the prep time (T) across the dishes currently on the menu, so each becomes a comparable number around zero:

$$z(x) = (x - \text{mean of } x) \div (\text{standard deviation of } x)$$

if every dish is the same (sd = 0), set z = 0

The two context fields stay as direct signals: **M** = promote level the owner sets (0 none, 1 promote, 2 strong); **W** = weather fit (+1 it matches today's weather, 0 neutral, -1 mismatch).

3. The two formulas

$$\text{Full — Score}_i = a \cdot z(V_i) + c \cdot M_i + d \cdot W_i - e \cdot z(T_i)$$

with weather and prep time

$$\text{Simple — Score}_i = a \cdot z(V_i) + c \cdot M_i$$

without weather and prep time

The signs carry the logic: + **value density** (more profit), + **promote** (owner's push), + **weather fit**, - **prep time** (slow dishes lower throughput and add labour). Suggested starting weights — **a = 1.0** (value), **c = 1.5** (promote, kept highest so the owner stays in control), **d = 0.8** (weather), **e = 0.5** (prep penalty). Tune them later (see §6).

4. Reading the output — and a 0–100% share

The score is centred at zero (roughly -3 to +3): positive = above average, promote it; negative = below average, leave it. Rank the dishes by score. To turn the scores into a **share** that sums to 100% — useful for weighting the AI's recommendation mix — pass them through softmax:

$$\text{share}_i = \exp(\text{Score}_i) \div [\exp(\text{Score}_1) + \exp(\text{Score}_2) + \dots + \exp(\text{Score}_n)]$$

5. Worked example

Four dishes, simple case (no promote, no weather). Value density and prep time are standardised across the four, then combined as Score = z(V) - 0.5·z(T); softmax gives the share.

Dish	Price	Qty	V = P÷Q	Prep	z(V)	z(T)	Score	Share
A	95	1	95	12	-0.14	-0.60	+0.16	26%
B	50	2	25	10	-1.41	-1.13	-0.84	9%
C	180	1	180	20	+1.41	+1.53	+0.64	42%
D	110	1	110	15	+0.14	+0.20	+0.04	23%

C wins (premium, even after a prep penalty); **B is negative**, so don't promote it. Now if the owner **promotes A** ($M = 1$): A's score becomes $0.16 + 1.5 = \mathbf{1.66}$, jumping to the top — the owner stays in control. On a **hot day** with D a cold drink ($W = +1$): D becomes $0.04 + 0.8 = \mathbf{0.84}$ and rises into contention.

6. Limits, and the upgrades worth knowing

- **This optimises revenue-density, not true profit** — because cost is left out. The moment a cost field is added, swap the value density to margin density: $V_i = (\text{Price}_i - \text{Cost}_i) \div \text{Quantity}_i$. That one change makes it genuinely profit-optimal; everything else stays the same.
- **It ignores demand.** Promoting a valuable dish nobody wants is a wasted slot. The real profit move is to push the high-margin, *under-selling* dishes. Once sales data exists, add a headroom term that favours dishes with room to grow: $+g \cdot (-z(\text{units sold}))$.
- **Recompute per branch and per time-of-day**, guard against $sd = 0$, and **learn the weights**: start with the suggested values, then check — did promoting the top-score dishes actually lift sales? — and nudge a, c, d, e . That turns a fixed formula into one that improves itself.

In short: value density carries the profit signal, the owner's promote and the weather refine it, prep time keeps the kitchen realistic, and the score tells the AI — in one positive-or-negative number — exactly which dish to push.